



FASR Science Case # WG1-4

# Revealing Electron Acceleration and Transport by Termination Shocks in Solar Flares with FASR

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## 1 Science Goal(s)

*Briefly summarize the key science goal(s) for this science case. A few sentences will be sufficient.*

Test whether and how flare termination shocks contribute to electron acceleration, and determine how subsequent trapping, escape, and precipitation shape the observed radio sources. FASR imaging spectroscopy will locate the shock relative to the reconnection outflow and loop-top trap, measure the evolving nonthermal electron population, and distinguish signatures of acceleration from those produced by transport through the flare magnetic geometry.

## 2 Scientific Rationale

### 2.1 Scientific Importance

*Provide a brief discussion on the scientific importance of this science case.*

Termination shocks (TSs) form where reconnection outflows encounter the closed flare arcade and may accelerate electrons in the loop-top region (Forbes 1986; Takasao et al. 2015). Simulations predict acceleration and trapping at such shocks, but observations have not yet established how often TSs form or how much they contribute to the flare electron population (Kong et al. 2019; Li et al. 2025). Radio sources at the shock and downstream loop top are compact, rapidly variable, and embedded in brighter flare emission. The case therefore requires frequency coverage that spans coherent shock tracers and gyrosynchrotron emission, together with imaging cadence and dynamic range sufficient to follow both components.

### 2.2 Uniqueness to FASR Capabilities

*Is this science case uniquely addressed by FASR? Why can't other facilities address this science and achieve the same goal?*

FASR can test the TS interpretation by measuring, in the same event:

- (1) the position, extent, and frequency structure of a shock candidate relative to the reconnection outflow and arcade;
- (2) the evolving spectra and polarization of the shock, loop-top, fan, and footpoint sources; and
- (3) timing and morphology changes that distinguish prompt acceleration at the shock from subsequent trapping, escape, and precipitation.

To distinguish acceleration at the termination shock from subsequent electron transport, we will compare the timing and spectral evolution of the shock-associated, loop-top, and footpoint sources. Emission appearing first near the shock and later along a loop would be consistent with acceleration followed by transport, but timing alone is insufficient: opacity, detection thresholds, and accumulation of electrons in a magnetic trap can also produce delayed radio emission. We will therefore interpret the timing together with spatially resolved radio spectra and polarization, supported by EUV and hard-X-ray observations. Radio magnetic-field diagnostics help constrain the environment through which electrons move, but do not directly identify where they gained energy.

## 2.3 Synergies

*Describe potential synergies/complementarities between this FASR science case and those from current/future/planned facilities at all wavelengths (e.g., DKIST, MUSE, FIERCE, COSMO, ngGONG, etc.).*

Hard X-ray imaging and spectroscopy constrain the energetic electrons in dense plasma. EUV and soft X-ray imaging and Doppler measurements locate reconnection outflows, downflows, and the flare arcade. Combining these data with FASR source positions and spectra allows the shock candidate, the trapped loop-top population, and precipitation at the footpoints to be compared in a common geometry.

## 2.4 Measurements Required by FASR

*Provide a description of the necessary measurements to be carried out by the FASR to adequately address this science case. Please coordinate these measurements with the Science Requirements table in Section III.*

FASR should obtain co-temporal Stokes I and V imaging spectroscopy of the above-the-loop-top system. The required observables are the shock candidate's position and split-band structure, the spectra of the loop-top and footpoint sources, and their relative onset and evolution. A one-second cadence tests the acceleration-versus-transport timing; a separate fine dynamic-spectrum product tests short-lived decimetric structure. Any claim about the accelerator will be made only after comparing these measurements with EUV and hard-X-ray source locations.

# III. Science Requirements Tables

## (A) Observational Target Description

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

The target is the connected above-the-loop-top system rather than an isolated radio pixel. Resolving the shock candidate together with the outflow, fan, loop-top trap, and arcade allows source motion and spectral evolution to be compared along the expected acceleration and transport path, as demonstrated by radio spectral imaging of flare termination-shock candidates (Chen et al. 2019; Luo et al. 2021). The compact shock tracer sets the angular-resolution and high-dynamic-range demands, while the surrounding fan and arcade set the field-of-view and largest-scale requirements.

| (A) OBSERVATIONAL TARGET                         |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type of observation<br>(what defines a ‘target’) | Triggered radio imaging spectroscopy of an eruptive flare with an above-the-loop-top TS candidate, including the shock surface, supra-arcade fan/downflows, loop-top trap, and flare arcade                                                                                                                                                                    |                                                                                                                                                                                      |
| Number of targets                                | 1 flare per observing sequence; accumulate all suitable eruptive M/X-class events for a statistical sample                                                                                                                                                                                                                                                     |                                                                                                                                                                                      |
| Size of a single target (arcsec x arcsec)        | TS/spike source $\sim 10''$ scale; full loop-top/fan/arcade complex typically $\sim 1'-2'$ across                                                                                                                                                                                                                                                              |                                                                                                                                                                                      |
| Distribution of all targets (arcmin x arcmin)    | Within a $5' \times 5'$ flare field of view centered on the above-the-loop-top region and underlying arcade                                                                                                                                                                                                                                                    |                                                                                                                                                                                      |
| Peak brightness (sfu/beam or Kelvin)             | Coherent dm bursts: integrated flux $10^3$ –several $10^4$ sfu; corresponding apparent peak $T_b \sim 6 \times 10^9$ –a few $10^{11}$ K for an illustrative $30''$ FWHM Gaussian image at 1.5 GHz (see D); TS/loop-top gyrosynchrotron continuum $T_b \sim 10^5$ – $10^7$ K at a few GHz for $B \sim 300$ G and $n_{\text{rel}} \sim 10^4$ – $10^6$ cm $^{-3}$ |                                                                                                                                                                                      |
| RMS brightness (sfu/beam or Kelvin)              | Recover the faintest accepted model component at $5\sigma$ in Stokes I and V; the numerical rms is determined by the source-injection test                                                                                                                                                                                                                     |                                                                                                                                                                                      |
| Expected circularly polarized flux density       | Stokes V<br>(sfu/beam)                                                                                                                                                                                                                                                                                                                                         | $\sim 10^{-3}$ – $10^{-1}$ sfu/beam for MK-level gyrosynchrotron continuum; coherent dm-burst values are event-dependent and may be reported as integrated flux rather than sfu/beam |
|                                                  | V/I                                                                                                                                                                                                                                                                                                                                                            | $\sim 5$ –50%, depending on frequency, viewing angle, optical depth, and emission mode                                                                                               |
| Expected linearly polarized flux density         | Stokes Q or U<br>(sfu/beam)                                                                                                                                                                                                                                                                                                                                    | N/A; Stokes Q and U are not required for this case                                                                                                                                   |
|                                                  | Q/I or U/I                                                                                                                                                                                                                                                                                                                                                     | N/A                                                                                                                                                                                  |

## (B) Spectro-Temporal Requirements Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

The 1–20 GHz span covers decimetric coherent shock signatures and the centimetric gyrosynchrotron continuum from electrons after acceleration. Frequencies near 1–3 GHz must preserve split-band lanes; the centimetric band constrains the spectrum of trapped and transported electrons. The routine product should sample the smooth spectrum at no worse than about 2% fractional spacing, while a dedicated low-frequency product retains 1–10 MHz channels when coherent structure is present. Approximately one-second imaging compares prompt and delayed emission, but timing alone cannot identify the accelerator because opacity, sensitivity, and trapping can create the same delay.

The same frequency span connects the two tests. For gyroresonance or low-harmonic gyrosynchrotron diagnostics,  $\nu_B = 2.8B$  MHz gives  $B \approx 357\nu_{\text{GHz}}/s$  G; this range samples fields from the weak above-the-loop-top region to stronger flare loops. These field estimates constrain the environment through which electrons move, while the source timing and spectral evolution provide the acceleration-versus-transport test.

The continuum and spike diagnostics have different spectral scales. In the split-band interpretation, the density compression is  $X = (\nu_{\text{high}}/\nu_{\text{low}})^2$  (Chen et al. 2019). At 1–3 GHz, 1–10 MHz channels keep the fractional frequency error below a few percent and avoid mixing adjacent lanes. No fixed number of continuum channels is a baseline; the design must provide enough samples to meet the fractional-spacing requirement and preserve the fine-structure product when triggered. Raw dynamic spectra may need order-1 MHz channels and approximately 10 ms integrations for individual bursts (Bastian et al. 2025).

| (B) SPECTRAL-TEMPORAL REQUIREMENTS      |                                                                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Central Frequency (GHz)                 | 1–20 GHz coverage; no single center frequency                                                                               |
| hline Instantaneous Bandwidth (GHz/pol) | Sufficient tuning or processing to obtain the full science span within each one-second cadence; implementation dependent    |
| hline Spectral resolution [MHz]         | $\Delta\nu/\nu \lesssim 2\%$ for the continuum; 1–10 MHz below 3 GHz when a split-band or fine-structure trigger is present |
| hline Temporal resolution (in seconds)  | $\sim 1$ s for routine imaging spectroscopy; approximately 0.01 s raw dynamic spectra when needed for dm fine structure     |

| (B) SPECTRAL-TEMPORAL REQUIREMENTS (continued) |  |
|------------------------------------------------|--|
| hline                                          |  |

### (C) Polarization Product Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

Stokes I gives the brightness-temperature spectrum. Stokes V is required for magnetic polarity and for the field/viewing-angle dependence of gyrosynchrotron or gyroresonance emission; it does not by itself determine the three-dimensional field or locate the accelerator. The baseline polarization test is recovery of the sign and spectral evolution of an injected V spectrum; a 2-percentage-point V/I error is a goal for bright sources. Calibrated total-power/zero-spacing information is required across the full 1–20 GHz science band and must be synchronized with the approximately 1 s imaging-spectroscopy products. It provides absolute flux and diffuse spectral information needed to interpret gyrosynchrotron and gyroresonance emission; combine it with interferometric data to recover source structure. At 2.8 GHz it also provides the F10.7 reference. A 5% flux-density scale is a working baseline for relative spectral fitting, with tighter accuracy treated as a goal.

| (C) POLARIZATION DATA PRODUCTS REQUIRED |          |
|-----------------------------------------|----------|
| y                                       | Stokes I |
| n                                       | Stokes Q |
| n                                       | Stokes U |
| y                                       | Stokes V |

### (D) Imaging Requirements Discussion

*Provide a brief discussion describing how these values are obtained/estimated, any trade-offs, interrelationships between the values, or anything else that is not captured in the following table.*

Fast reconnection outflows in eruptive flares should slow abruptly when they run into the closed flare arcade. Line-tied reconnection models and flare-loop simulations place fast-mode shocks in this above-the-loop-top region, where the downward jet, the newly closed arcade, and the hot supra-arcade plasma meet (Forbes 1986; Takasao et al. 2015). VLA imaging spectroscopy has found dynamic, surface-like decimetric spike sources at the end points of fast downflows, and the split-band structure in one event is consistent with emission from the upstream and downstream sides of a shock front (Chen et al. 2015; Chen et al. 2019). A later VLA event placed a similar spike source about 60 Mm above a flare arcade, embedded in a supra-arcade fan with many plasma downflows (Luo et al. 2021). The source volume to be imaged includes the shock surface, reconnection outflow, fan/downflow system, loop-top trap, and the underlying flare arcade. Recent multiwavelength modeling of an above-the-loop-top magnetic bottle gives the same geometry: a compact hard-X-ray source and an extended microwave source lie above the arcade, where electrons are trapped and accelerated with help from a fast-mode termination shock (Chen et al. 2024).

The observed TS candidates are small compared with the full flare system. The 2012 March 10 VLA event showed 1.0–1.6 GHz spike centroids forming an extended structure of order  $10''$ , located about 60 Mm above the flare arcade (Luo et al. 2021). In the split-band event, centroids from the two

lanes outlined nearly co-spatial surfaces separated by about 0.8 Mm, or only about  $1''$  on the Sun (Chen et al. 2019). The science requirement is to distinguish the approximately  $10''$ -scale shock candidate from the arcade and fan and to recover frequency-dependent centroids; source-injection tests should establish whether a given event supports about  $1''$  centroid precision. A  $5' \times 5'$  field covers the shock, loop-top trap, fan, downflow tracks, and arcade, while recovery of scales of at least  $300''$  preserves the diffuse context.

Bright coherent or compact flare sources can sit in the same field as much weaker extended continuum emission. Coherent dm-wavelength bursts can reach thousands to several  $10^4$  sfu, and rare events can be still brighter; typical observed dm-wavelength source sizes are tens of arcseconds once coronal scattering is included (Bastian et al. 2025). For a Gaussian source image, the Rayleigh–Jeans conversion gives the apparent peak brightness

$$T_b = 1.222 \times 10^{10} \frac{S_{\text{int}}/\text{sfu}}{(\nu/\text{GHz})^2(\theta_{\text{maj}}/\text{arcsec})(\theta_{\text{min}}/\text{arcsec})} \text{ K},$$

where the FWHM axes describe the observed image, including scattering and beam broadening (NRAO, Flux Density–Brightness Temperature Conversion). Assigning the integrated burst flux to a circular Gaussian of illustrative FWHM  $30''$  at 1.5 GHz gives  $T_b \simeq 6 \times 10^9$  K per  $10^3$  sfu. This conditional conversion supplies the Table A brightness range; it is not an independent measured range or a typical termination-shock brightness. A different angular extent or frequency changes the conversion, and unresolved components can have higher intrinsic brightness.

The faint emission of interest is the extended gyrosynchrotron or free-free component that marks where accelerated electrons are trapped, leak out, or encounter denser plasma. Rather than prescribe an instrument noise temperature from the current array, the baseline acceptance test injects the faintest physically plausible component from the particle-transport models into representative FASR data. The image must recover that component at  $5\sigma$ , preserve its spectral slope to 10%, and keep the shock/looptop morphology separable from the bright coherent source. The resulting Kelvin rms and calibration margin are design-phase outputs; the 1000 : 1 dynamic-range target remains because it is tied to the bright-to-faint source contrast.

| <b>(D) IMAGING REQUIREMENTS</b>                                 |                                                                                                                                          |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Required angular resolution (arcsec)<br>(single value or range) | Resolve the approximately $10''$ -scale shock candidate from the arcade and fan; validate centroid precision with source-injection tests |
| hline Largest angular scale required (arcsec)                   | $\geq 300''$                                                                                                                             |
| Mapped image size (arcmin x arcmin)                             | $5' \times 5'$                                                                                                                           |
| Required pixel resolution (arcsec)                              | Sample the synthesized beam with several pixels at each frequency                                                                        |

| (D) IMAGING REQUIREMENTS (continued)                                                                      |                                                                                                                                                                                                                                               |        |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| hline Number of output/image channels                                                                     | Enough samples to meet the $\Delta\nu/\nu \lesssim 2\%$ continuum requirement, plus triggered fine-structure windows; no fixed count is a baseline                                                                                            |        |
| hline Output bandwidth (minimum and maximum frequency - GHz)                                              | 1–20 GHz                                                                                                                                                                                                                                      |        |
| Channel width (MHz)                                                                                       | Set by the fractional-spacing requirement for the continuum; 1–10 MHz below 3 GHz for split-band imaging                                                                                                                                      |        |
| hline Required rms (sfu/beam or Kelvin) [per channel] (if polarization products required define for each) | Recover the faintest accepted model component at $5\sigma$ in Stokes I and V; the Kelvin value is set by source-injection tests                                                                                                               |        |
| hline Dynamic range within image (if polarization products required define for each)                      | $\geq 1000 : 1$ in Stokes I, validated against bright coherent and faint continuum components                                                                                                                                                 |        |
| hline Polarization accuracy (%)                                                                           | Recover the sign and spectral evolution of Stokes V; 2 percentage points in V/I is a goal for bright sources                                                                                                                                  |        |
| hline Zero spacing/total power required?                                                                  | Yes. Across 1–20 GHz, synchronized with the approximately 1 s imaging products; retain matching total-power spectra for any 0.01 s dynamic-spectrum windows. Combine with interferometric data to recover absolute flux and diffuse emission. |        |
| Required maximum latency (in seconds, or N/A)                                                             | N/A for offline science analysis                                                                                                                                                                                                              |        |
| Required flux density scale calibration accuracy                                                          |                                                                                                                                                                                                                                               | 1-3%   |
|                                                                                                           | x                                                                                                                                                                                                                                             | 5%     |
|                                                                                                           |                                                                                                                                                                                                                                               | 10%    |
|                                                                                                           |                                                                                                                                                                                                                                               | 20-50% |

## 2.5 Other Performance or Functional Requirements

*If there are any additional performance or functional requirements not captured above, describe them here. For example, beamforming array mode, phased array, etc.*

## 3 Appendix

*Please provide any other relevant material necessary to understand and substantiate this Science Case.*

### Physical-model benchmarks and interpretation limits

The following material records the physical basis for the spectral and brightness estimates used in Tables A–D. It is retained as quantitative context rather than as a separate observing requirement.

The radio images must separate density structure at the shock from the trapped nonthermal electrons that radiate after acceleration. Coherent plasma emission from stochastic spikes traces density fluctuations near the shock, with  $\nu \approx s\nu_p$  and  $\nu_p \approx 8980\sqrt{n_e}$  Hz; emission at 1–2 GHz corresponds to  $n_e \sim 10^{10}–5 \times 10^{10} \text{ cm}^{-3}$  for fundamental emission, or four times lower for second-harmonic emission. Gyrosynchrotron emission from mildly relativistic electrons gives the broadband cm-wavelength spectrum used to infer the electron energy distribution, magnetic field, viewing angle, and source depth. Thermal free-free emission measures the hot background plasma, and gyroresonance or low-harmonic gyrosynchrotron emission can appear in stronger-field loops below the shock (Dulk 1985; Bastian, Benz, and Gary 1998; Nindos 2020; Gary 2023). Particle-transport calculations connect these radio quantities to the acceleration physics: TS-driven acceleration and magnetic trapping can place hundreds-of-keV to MeV electrons in the loop-top region, with microwave and hard-X-ray source morphologies that depend on transport as well as acceleration (Kong et al. 2019; Kong et al. 2022; Li et al. 2022; Li et al. 2025).

The Li et al. simulations set the scale for the expected incoherent component. In the 2017 September 10 event model, the field is about 100 G in the current sheet and a few to several hundred gauss in compressed islands or island-merging regions, while electrons reach a few hundred keV to several MeV and the  $> 300$  keV population is concentrated near the loop top (Li et al. 2022). The 3D flare calculation with an explicit TS accelerates electrons above 100 keV within the first  $\sim 30$  s, reaches energies above 2 MeV, and gives a TS-region power-law index near 4.9. For a 20 MK seed population, the number above 100 keV is of order  $10^{-5}$  of the current-sheet/loop-top electrons, comparable in fractional abundance to the microwave-inferred  $> 300$  keV densities of  $10^4–10^5 \text{ cm}^{-3}$  in the TS/loop-top region reported from EOVSAs (Chen et al. 2020; Li et al. 2025). These thresholds are not interchangeable: the simulation fraction above 100 keV and the EOVSAs density above 300 keV refer to different populations and normalizations, so their numerical similarity is only a scale comparison. In fields of 100–300 G, 5–20 GHz samples harmonic numbers  $s \sim 6–70$ , while 3 GHz reaches the low-harmonic/turnover part of stronger-field sources.

The modeled brightness ranges define the source-injection cases. Their recoverability must be tested by inserting the corresponding particle-transport sources into representative FASR visibilities and measuring the recovered rms, dynamic range, and spectral bias at each frequency. A source is accepted when the faint component is recovered at  $5\sigma$  and its fitted spectrum is accurate to 10%; no fixed image-noise temperature is assumed before that test.

The Dulk–Marsh estimates above depend on the adopted low-energy cutoff and distribution normalization. The effective  $n_{\text{rel}}$  is degenerate with  $B$ , source depth, and spectral index; the simulation fraction above 100 keV cannot be inserted directly as the EOVSA density above 300 keV without converting the energy distributions and normalizations. The estimates therefore indicate plausible brightness scales rather than a direct 10-GHz magnetic-field measurement.

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